

Ultrapowers

If (X, d) is a metric space, we may consider the space $\ell^\infty(X)$ to be the space of all bounded sequences in X with metric defined by

$$d_\infty(x, y) = \sup_{n \in \mathbb{N}} d(x_n, y_n).$$

If ω is an ultrafilter on $\mathbb{N}$, then we define an equivalence relation $x \sim y$ on $\ell^\infty(X)$ by saying they are equal if

$$\lim_{\omega} d(x_n, y_n) = 0.$$

Then, we define the equivalence classes

$$[x]_{\omega} = \{y \in \ell^\infty(X) \mid y \sim x\}$$

and take the quotient $X_{\omega} = \ell^\infty(X)/\sim$, with distance metric defined by

$$d_{\omega}([x]_{\omega}, [y]_{\omega}) = \lim_{\omega} d(x_n, y_n).$$

This space is known as the *ultrapower* of the metric space (X, d) . The map $\iota_{\omega}: X \rightarrow X_{\omega}$ defined by $x \mapsto [(x, x, \dots)]_{\omega}$ is an isometric embedding of (X, d) into (X_{ω}, d_{ω}) .

Generally, we assume that the ultrafilter ω is nonprincipal, since if ω is principal, then taking the limit along the ultrafilter is equivalent to extracting the element of $\mathbb{N}$ that defines the principal ultrafilter.

A useful fact is that if (X, d) is complete, then so too is (X_{ω}, d_{ω}) .

Now, suppose G is a group, and let $d: G \times G \rightarrow [0, \infty)$ be a metric on G . We say the metric d is left (or right) invariant if $d(hg_1, hg_2) = d(g_1, g_2)$ for left invariance and $d(g_1h, g_2h) = d(g_1, g_2)$ for right invariance. A metric that is both left and right invariant is called bi-invariant, and the space (G, d) is called a metric group.

Example: Let G be a finitely generated group, and let $S \subseteq G$ be a finite generating subset. Then, (G, d_S) is a left metric group, where d_S denotes the word metric — i.e., $d_S(g, h)$ is the minimum word length necessary to construct the element gh^{-1} from S .

In the next section, we will discuss two important metric groups and various important properties thereof.

Sofic Groups

Suppose S_n is the group of permutations of the set $[n] = \{0, \dots, n-1\}$. Then, the *Hamming length* of a permutation $\sigma \in S_n$ is given by

$$\ell(\sigma) = \frac{1}{n} |\{i \in [n] : \sigma(i) \neq i\}|.$$

The metric $d: S_n \times S_n \rightarrow [0, \infty)$ given by $d(x, y) = \ell(xy^{-1})$ then is a bi-invariant length function.¹

Definition: A countable discrete group Γ is *sofic* if for every $\varepsilon > 0$ and every finite subset F of $\Gamma \setminus \{e\}$, there is n and a map $\Phi: \Gamma \rightarrow S_n$ such that $\Phi(e) = \text{id}$, and for every $g, h \in F \setminus \{e\}$, we have

- $d(\Phi(gh), \Phi(g)\Phi(h)) < \varepsilon$;
- $\ell(\Phi(g)) > r(g)$, where $r(g)$ is a constant depending only on g .

¹For invariance, recall that conjugation preserves cycle type, which is essentially what the Hamming distance is measuring. One can show that a metric being bi-invariant is equivalent to it being preserved under conjugation.

One of the main benefits of the ultrapower method is that it allows for a conversion from this local definition to a global definition. Fixing a free ultrafilter ω on $\mathbb{N}$, we define the ultrapower

$$\prod_{\omega} S_n := \prod_{n \in \mathbb{N}} S_n / \left\{ x \in \prod_{n \in \mathbb{N}} S_n : \lim_{\omega} \ell_{S_n}(x) = 0 \right\}.$$

It follows from this definition that Γ is sofic if and only if there exists an injective homomorphism $\Phi: \Gamma \rightarrow \prod_{\omega} S_n$ for any free ultrafilter ω on $\mathbb{N}$.

Definition: Suppose G and H are groups with invariant length functions, F a subset of G , and ε is a positive number. A function $\Phi: G \rightarrow H$ is called an (F, ε) -approximate homomorphism if $\Phi(e_G) = e_H$ and for every $g, h \in F$,

- $\left| \ell_H(\Phi(gh)\Phi(h)^{-1}\Phi(g)^{-1}) \right| < \varepsilon$;
- $|\ell_H(\Phi(g)) - \ell_G(g)| < \varepsilon$.

Theorem: A countable discrete group Γ is sofic if and only if for every $\varepsilon > 0$ and every finite $F \subseteq \Gamma$, there is n and an (F, ε) -approximate homomorphism from Γ to S_n , where Γ is regarded as an invariant length group with respect to the trivial invariant length function.

Here, the trivial invariant length function refers to the function

$$\ell_0(x) = \begin{cases} 0 & x = e_{\Gamma} \\ 1 & \text{else.} \end{cases}$$

Local Embeddability

We discuss a more general notion than soficity that we will specialize to the sofic case.

Definition: Let G and C be groups. If $K \subseteq G$ is finite, a map $\varphi: G \rightarrow C$ is called a K -almost homomorphism of G into C if it satisfies

- $\varphi(k_1 k_2) = \varphi(k_1)\varphi(k_2)$ for all $k_1, k_2 \in K$;
- $\varphi|_K: K \rightarrow C$ is injective.

We call a collection $\mathcal{C}$ a *class of groups* if it is a collection of groups such that if $C \in \mathcal{C}$ is a member, and $C' \cong C$ are isomorphic, then $C' \in \mathcal{C}$. The main ones we will be interested in are the class of finite groups and the class of amenable groups.

Definition: If $\mathcal{C}$ is a class of groups, then we say G is *locally embeddable* into $\mathcal{C}$ if for every finite subset $K \subseteq G$, there exists a group $C \in \mathcal{C}$ and a K -almost homomorphism $\varphi: G \rightarrow C$.

Proposition: Let $\mathcal{C}$ be a class of groups. Every subgroup of a group that is locally embeddable into $\mathcal{C}$ is locally embeddable into $\mathcal{C}$.

Proof. Let G be a group that is locally embeddable into $\mathcal{C}$, and let H be a subgroup of G . Given a finite subset $K \subseteq H$, then $\varphi: G \rightarrow C$ is a K -almost homomorphism into $C \in \mathcal{C}$ such that $\varphi|_H: H \rightarrow C$ is a K -almost homomorphism of H into C , so H is locally embeddable into $\mathcal{C}$. $\square$

Proposition: Let G be a group, and let $\mathcal{C}$ be a class of groups. Then, G is locally embeddable into $\mathcal{C}$ if and only if every finitely generated subgroup of G is locally embeddable into $\mathcal{C}$.

Proof. One direction follows from the previous proposition.

Now, suppose every finitely generated subgroup of G is locally embeddable into $\mathcal{C}$. Let $K \subseteq G$ be a finite subset, and let $H = \langle K \rangle$ be the subgroup generated by K . Then, since H is locally embeddable into $\mathcal{C}$, there is a K -almost homomorphism $\varphi: H \rightarrow C$ of H into $C \in \mathcal{C}$. We may extend φ to G arbitrarily by setting $\varphi(g) = e_C$ for all $g \in G \setminus H$, whence $\varphi: G \rightarrow C$ is a K -almost homomorphism of G into C . $\square$

Examples and Inheritance Properties

Proposition: Every finite group is sofic.

Proof. Let G be a finite group. By Cayley's Theorem, we have a map $L: G \rightarrow \text{Sym}(G)$ defined by $L(g)(h) = gh$ for all $g, h \in G$. Since L is a homomorphism, we have that $\ell_G(L(g_1g_2)L(g_2)^{-1}L(g_1)^{-1}) = 0$. Similarly, since $\ell_G(L(g)) = \ell_G(g)$, it follows that this is a (G, ε) -approximate homomorphism. $\square$

Proposition: Amenable groups are sofic.

Proof. Let Γ be a amenable countable discrete group, F a finite subset of Γ , and ε a positive real number. Find a $(F \cup F^{-1}, \varepsilon)$ -invariant subset K of Γ , which follows from the Følner condition. If $\gamma \in F$, then define

$$\sigma_\gamma(x) = \gamma x$$

for each $x \in \gamma^{-1}K \cap K$, and extend σ_γ to an arbitrary permutation of K . Then, this defines a $(F, 2\varepsilon)$ -approximate homomorphism to the group of permutations of K endowed with the Hamming length. Therefore, Γ is sofic. $\square$

Hyperlinear Groups

References

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